

Controls & BMS Spec Sheet

Doc No: HVAC-003

Rev: A

Date: 2026-05-12

Project: Lakefront Office Tower — Phase II

ACME Engineering Associates, Inc.
123 Industrial Pkwy, Suite 400
Chicago, IL 60601

SECTION 23 09 23 — DIRECT DIGITAL CONTROLS FOR HVAC

1.0 SCOPE OF WORK

Furnish, install, program, and commission a fully integrated BMS providing Direct Digital Control (DDC) of all HVAC systems. BMS shall interface with electrical, fire alarm, security, and lighting via open BACnet/IP protocol. Niagara N4 supervisor platform required.

2.0 EQUIPMENT SCHEDULE

Controller Tag	System Served	I/O Points	Protocol	Qty
AHU-DDC-1	AHU-1 (Low Floors)	38 pts	BACnet/IP	1
AHU-DDC-2	AHU-2 (High Floors)	38 pts	BACnet/IP	1
AHU-DDC-3	AHU-3 & AHU-4	52 pts	BACnet/IP	1
VAV-FLR-DDC	VAV Boxes Floors 3–9	8 pts ea	BACnet MS/TP	42
VAV-FLR-DDC	VAV Boxes Floors 10–18	8 pts ea	BACnet MS/TP	39
CHP-DDC	Chiller Plant	64 pts	BACnet/IP	1
BMS-GW	Gateway / Integrations	N/A	BACnet/IP	1

3.0 SYSTEM SCHEMATIC / DRAWING

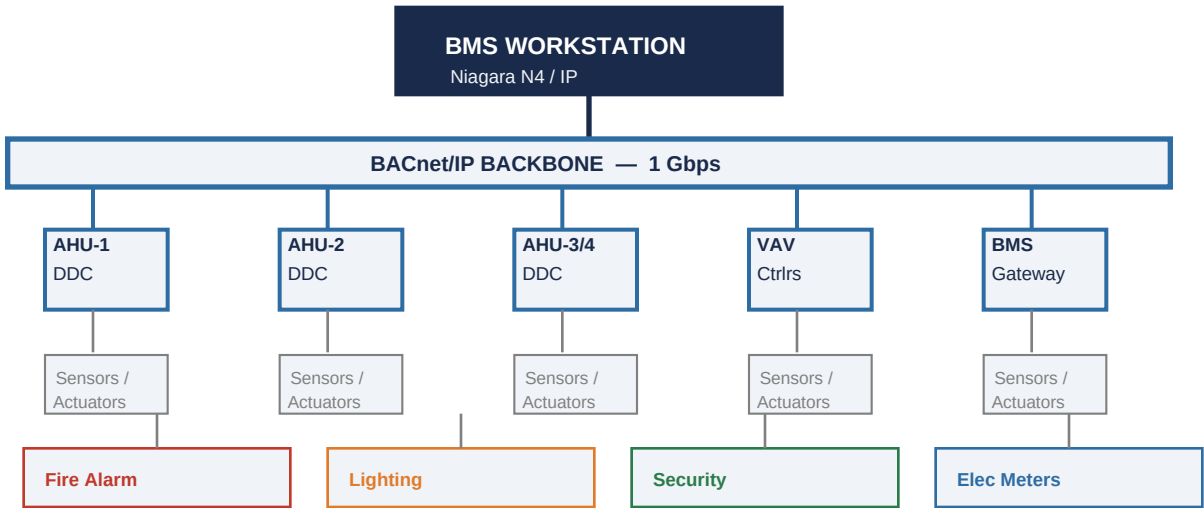


Figure 3 — BMS Network Architecture (Schematic — Not to Scale)

4.0 TECHNICAL SPECIFICATIONS

4.1 Supply Air Temperature Reset Schedule

OAT ≥ 75°F

SAT Setpoint = 55°F

OAT = 65°F	SAT Setpoint = 57°F
OAT = 55°F	SAT Setpoint = 60°F
OAT ≤ 45°F	SAT Setpoint = 65°F
Standard	ASHRAE 90.1-2022 Section 6.5.2.1

4.2 Static Pressure Reset & Economizer

SP Reset Range	1.5 in. w.g. (design) down to 0.5 in. w.g. (minimum)
SP Reset Trigger	When all VAV boxes <90% open; 0.1 in. per 5-min interval
Economizer Type	Differential enthalpy, high-limit shutoff OAT >75°F
OA Min. Flow	Demand-controlled ventilation per ASHRAE 62.1-2022
Standard	ASHRAE 90.1-2022 §6.5.3.2

5.0 INSTALLATION NOTES

- BMS contractor to provide graphics for all AHUs, VAV zones, chiller plant, and alarms.
- All controllers: battery-backed clock, flash memory — retain program on power loss.
- BACnet: BTL-listed devices only; provide interoperability test report.
- Provide 100% factory testing of all DDC panels prior to shipment.
- Commissioning per ASHRAE Guideline 1.5; functional performance tests required.
- BMS warranty: 2 years parts and labor; include software updates for 5 years.